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CAPSTONE ABSTRACT

Estimating the Impact of COVID-19 on ELA Student Achievement in New York City Public Schools

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The COVID-19 pandemic resulted in significant learning loss for K-12 children. A significant amount of research has been conducted on why learning loss occurred during and after the pandemic, but a lack of data and resources results in limited information in quantifying learning loss, especially for younger children. This research aims to explore achievement gaps in the post-pandemic period by comparing impoverished schools to non-impoverished schools using English Language Arts standardized test scores in New York City public schools for grades 3-8. This project uses two-way fixed effects models by using the poverty rate of individual schools as a treatment. Two models are featured, with the initial model applying a binary poverty classification treatment (high/low) and the second using continuous baseline poverty rates from the pre-period. Models were done separately for each grade level and results were verified using a sensitivity analysis to assess the parallel trends assumption. The models found that third and fourth graders in impoverished schools were affected the most relative to peers in lower-poverty schools, with test scores decreasing in the post-pandemic period respective to pre-pandemic poverty rates. This work aims to inform education policy decisions, intervention approaches, and equitable resource allocation by identifying how achievement gaps are broadened with school-level poverty.

1. Introduction

The COVID-19 pandemic resulted in significant learning loss for K-12 children. A vast amount of research has been conducted on why learning loss occurred during and after the pandemic, but a lack of data and resources results in limited information in quantifying learning loss, especially for younger children. The subject of English Language Arts (ELA) is the most crucial subject for understanding achievement gaps due to the necessary grasp of reading comprehension across all other subjects.

In combination with the circumstances of learning during the pandemic, children from disadvantaged backgrounds are said to have suffered the most during and post-COVID. Low-income households and communities did not have similar access to a stable internet connection, computers, or a dedicated space for studying compared to students from higher income households (Habashi et al.). Former research mainly quantifies the overall effects for K-12 children but often fails to consider the variety of socioeconomic backgrounds that may contribute to student success.

The motivation for this project stems from a need for informed education policy and a concern for achievement gaps that have not recovered in the post-pandemic period. Despite the large amount of pandemic-related literature, there remains a gap in quantifying learning loss in specifically for impoverished children in school systems across the United States. Existing research also fails to consider the age-related effects and the challenges that children in developmental stages face.

Unfortunately, data surrounding the pandemic is limited due to the worldwide shutdown. Education data is especially difficult to analyze considering changes in testing formats, inconsistent attendance rates, and disruptions in the classroom. Nevertheless, this research aimed to use the best possible data and practices needed to form concrete conclusions on the effect of the pandemic on learning loss. Using statistical modeling, this research aims to identify and explore suspected learning loss for children in grades three through eight using ELA standardized test scores in New York City in hopes of informing policy decisions.

2. Methods

2.1 Data Structure & Description

Data was provided by the New York City Department of Education OpenData. The main source of data provided information at the citywide, borough, district, and school levels from 2013 to 2023. Unfortunately, standardized test score scaling changed from 2017 to 2018, as well as from 2022 to 2023. The most comparable years were 2018 to 2022, with 2020 and 2021 excluded due to the cancellation of testing. Other testing information outside of 2018 to 2022 was removed for analysis. When evaluating test scores, the New York City Department of Education suppressed scores for certain grade levels who had under five test takers and absorbed those values into the “All Grades” category. These suppressed rows were removed and not incorporated into the final model. This project specifically required the use of school-level information, using 1121 unique schools in New York City. Grade level observations were balanced amongst elementary (grades 3-5) and middle (grades 6-8) school students, allowing for fair comparisons (See Appendix, Table 1A). The most essential variables in this data were the school identification number, the year, grade level, mean score of each school, and the number of students tested. Summary statistics of these variables are available in the Appendix in Table 2A.

The second data source was also provided by the New York City Department of

Education OpenData. This information provided a demographic snapshot of individual schools from 2017 to 2022, with information about racial and ethnic backgrounds, gender, and economic class. The most essential variables in this data were poverty rate and the Economic Need Index, which were both used to explore the best way to determine economic status within individual schools (See Appendix, Table 2A for summary statistics). This data was merged with the ELA comprehension data using the school identification number and year. Ultimately, the poverty rate was deemed the most appropriate metric for the project scope. Initially, the poverty rate for each year (2018, 2019, and 2022) were used. In order to provide consistency and reduce bias, a baseline year was used, with 2019 serving as the reference year for the poverty rate in individual schools. There was no missing information in this data, allowing for simple inspection and incorporation into the initial data source.

2.2 Exploratory Data Analysis

Exploratory data analysis was performed using cleaned and merged data. *Figure 1* displays the relationship between school poverty rates and ELA achievement, as indicated by mean test scores, for the year 2019. In the pre-period, high percentages of poverty within a school indicate lower test scores.

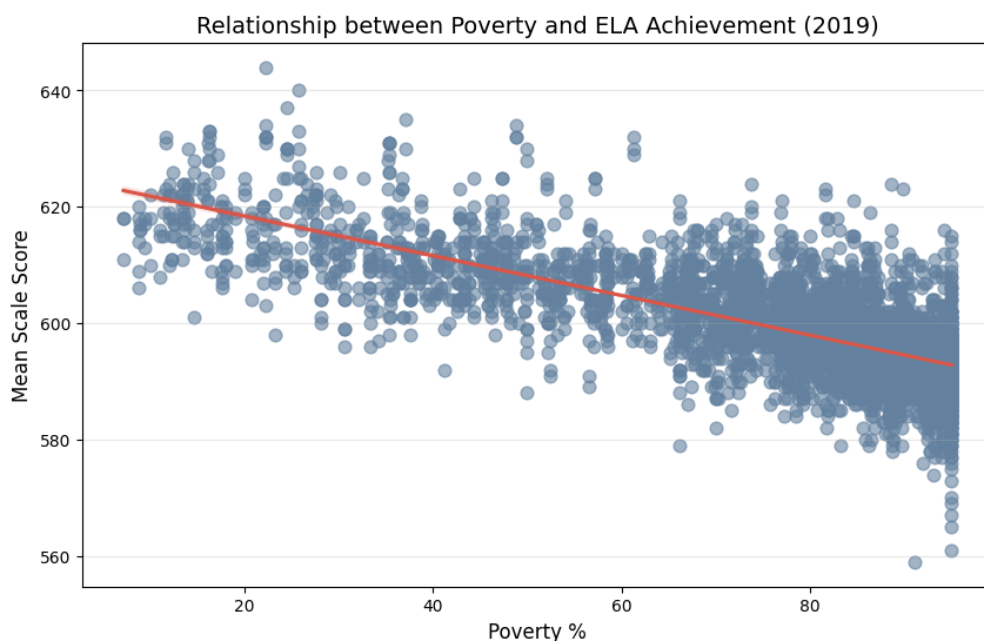


Figure 1: Relationship between Poverty & ELA Scores in 2019

This result aligns with former research surrounding the pandemic, as well as the significant findings on the effect of low-income schools on overall achievement. Previous research by Andy Hegedus highlights how school poverty is an even bigger indicator of student performance compared to individual student poverty levels, suggesting that students at an already impoverished school are at a systematic disadvantage regardless of individual economic background and effort.

Further analysis was conducted at the grade level. *Figure 2* displays the mean score across the years for each grade level. The continuous poverty variable was converted into a binary variable, where schools with a 60% or higher poverty rate were classified as a high

poverty school and those under 60% are classified as low poverty. This categorization follows Title I standards, where schools with a poverty rate that is 60% or higher are eligible for federal assistance.

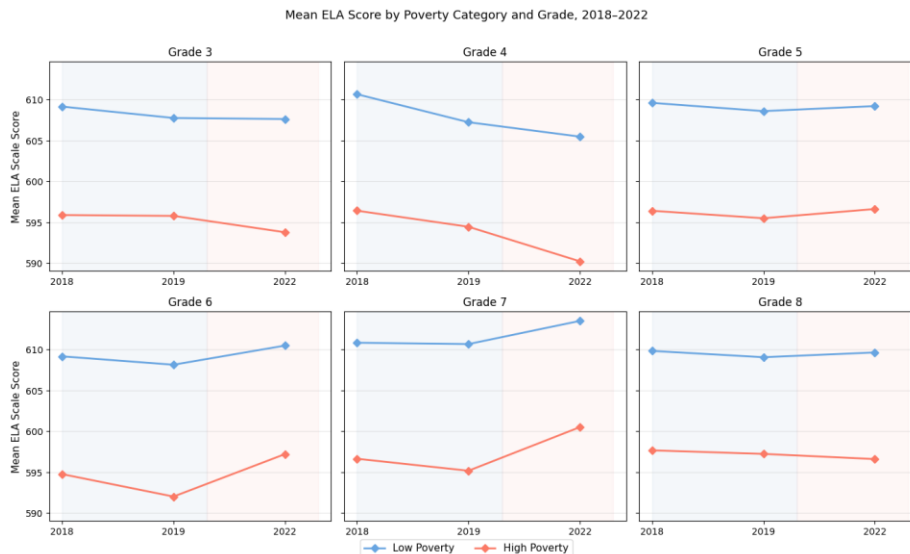


Figure 2: Mean ELA Test Score for High & Low Poverty Schools

This preliminary assessment gives insight into the future development of modeling. Grades 3 and 4 show sharp declines in mean ELA test scores among schools categorized as high poverty. Grades 5 through 7 yield unexpected, yet interesting, findings, where there is significant improvement across all schools. Grade 8 also shows that there is an achievement gap between high and low poverty schools, yet the difference is very slight.

In assessing district-level information, the data revealed that poverty rates vary within school districts, making it difficult to fully understand the effects of poverty and the pandemic. Recovery and effects varied a lot throughout New York City, making it difficult to do a district-level assessment due to the flattening of variation within individual schools (See Appendix, Figure 1A). Due to these findings, modeling was done with individual schools to allow for more careful analysis.

2.3 Modeling Approach

The initial modeling approach consisted of using a two-way fixed effect difference-in-difference (TWFE DID) model to assess the change in mean ELA test scores from 2019 to 2022. The treatment for this model was the poverty rate for individual schools. This metric was initially binary, where schools with a poverty rate of greater than 60% would be classified as high poverty schools and vice versa. This method did not fully encapsulate the overall effects of poverty within schools and was ultimately transitioned into a continuous treatment variable. A model was performed for each grade, 3 through 8, to fully understand and investigate the effects that COVID had on a variety of ages. The number of students tested was incorporated as a weight for all models.

The initial model used poverty as a binary treatment variable, where high poverty schools were classified as 1, and the remaining classified as 0. Finding the appropriate cutoff for high and low poverty consisted of exploring different ways of classification, such as comparison to the New York City median poverty rate, Title I eligibility, and the New York City Department of Education's Economic Need Index.

The model was performed using Python's linearmodels package with the PanelOLS function. Modeling was performed using a baseline poverty rate using information from 2019 to allow for consistent metrics and avoid post-COVID shocks in the poverty rate. Using the baseline poverty rate, as well as fixed effects for individual schools and time, the preliminary model simply focused on looking at the difference in post-COVID effects. The following equation represents the initial modeling process:

$$Mean\ Scale\ Score_{i,t} = \alpha_i + \alpha_t + \beta_1(Post_t * Treated_i) + \epsilon_{i,t}$$

Equation 1: Initial Model

This equation uses Mean Scale Score for some individual school, i , in period, t . The fixed effects are represented by α_i and α_t , which control for unobserved school characteristics and time-specific shocks, respectively. The β_1 coefficient is the metric of interest, which captures the differential effects in the post-COVID period for high poverty schools relative to the pre-COVID period. Initially, β_1 used the poverty rate for the corresponding year. It was ultimately decided that the use of a poverty rate baseline value to avoid post-COVID shocks in the poverty rate would allow the model to be more consistent.

While the preliminary plan was to evaluate the difference between test scores from 2019 to 2022, it was determined that the incorporation of one more pre-period year would assist in identifying pre-existing trends and identifying model bias. The incorporation of the year 2018 created a more complex model and required more assumptions to be met. The TWFE DID model requires the validation of the parallel trends assumption. Essentially, the parallel trends assumption states that in the absence of treatment, treatment and control groups would trend the same way over time (Columbia University). The parallel trends assumption is difficult to quantify, especially with a limited number of years of information. The typical way of identifying if the parallel trends assumption is met is by doing a placebo test using only the pre-period. In this case, 2018 would be the false pre-period and 2019 would be the false post-period. The goal with this assessment is to identify pre-existing trends in test scores and evaluate if these trends influence the accuracy of the full model.

Since the parallel trends assumption is difficult to validate, an alternative way to address this assumption was performed. Rambachan and Roth described the difficulty of validating the parallel trends assumption, finding that parallel trends are almost never perfectly true when using real-world data. Rather than attempting to perfectly validate this assumption, these researchers suggest that the magnitude of the assumption should be assessed and how much of a violation in parallel trends can impact results. Essentially, instead of viewing the parallel trends assumption as a pass or failure, this method investigates the sensitivity of the model. This framework uses the size of pre-period trend differences between groups to compare to the post-period trend differences. This framework uses the size of pre-trend differences to work as a benchmark to identify post-period violations. The sensitivity parameter, represented by the magnitude M , begins at 0, where no violation is allowed and parallel trends must be perfect. The magnitude ranges upwards to identify the point on which the finding no longer holds. In the scope of this project, with one pre-period difference benchmark (2018-2019) and one post-period year, this robustness check assesses whether the post-period coefficient is plausible considering potential violations of parallel trends.

The discovery of the HonestDiD robustness check resulted in model changes. While the initial model only had a post-COVID coefficient (β_1), the new model required both a pre-COVID coefficient and post-COVID coefficient. In this phase of modeling, the binary poverty

variable had been transitioned to a continuous treatment variable. This version of the model is like an event-study, rather than traditional difference-in-differences. The updated model similarly uses poverty rates from 2019 as a baseline for treatment. The model equation is as follows:

$$Mean\ Scale\ Score_{i,t} = \alpha_i + \alpha_t + \beta_1(Pre * Poverty)_{i,t} + \beta_2(Post * Poverty)_{i,t} + \epsilon_{i,t}$$

Equation 2: Improved Model

Similarly, the fixed effects are represented by α_i and α_t , for school and time fixed effects. The interaction coefficients are calculated as follows:

$$\beta_1 = (Poverty_{i,2019} * Pre_{i,2018})$$

$$\beta_2 = (Poverty_{i,2019} * Post_{i,2022})$$

Each grade is assessed with its own individual model. Since all modeling was performed in Python, coefficients and variance-covariance matrices for each model were exported from Python to be analyzed in R using the HonestDiD robustness evaluation. Coefficients were then analyzed based on the magnitude of parallel trends assumption violations and interpreted accordingly.

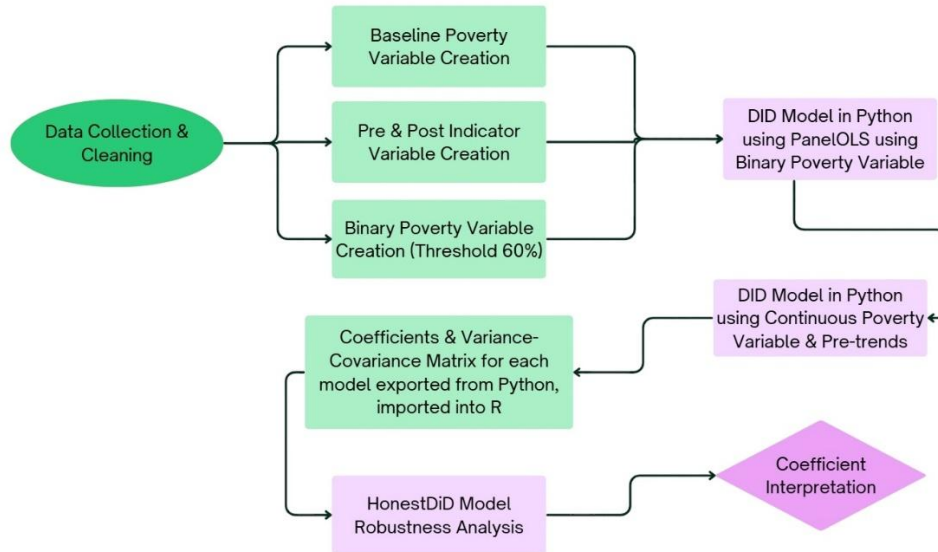


Figure 3: Complete Project Pipeline

A full pipeline of the project process is outlined in *Figure 3*, including data cleaning, feature engineering, construction of the model, and analysis of coefficients.

3. Modeling & Results

3.1. Initial Model

The preliminary model analyzed the changes in ELA test score from 2019 to 2022 in impoverished schools by controlling school and time fixed effects. Due to the simplicity of the model, a parallel trends assumption was not necessary. However, this is not to say that there is no

bias in the preliminary model.

Treatment was assigned using a binary poverty variable, where high poverty schools were classified as 1 and low poverty schools were classified as 0. The method of classification stems from Title I eligibility for federal assistance, which is 60% (Elsen-Rooney). Individual models were created for grades 3 through 8 to identify differences in education level. The equation for this model can be referenced in Section 2.3 as *Equation 1*.

Grade	DID Coefficient (β_1)	P-value
3	-2.4162	0.0001
4	-2.9672	0.0000
5	0.9056	0.1128
6	3.7025	0.0002
7	3.2393	0.0043
8	0.2513	0.7625

Table 1: Initial Model Results

The model performs as expected, with younger grades in impoverished schools experiencing the most significant learning loss. Almost all coefficients are significant in analyzing deeper learning loss for elementary and middle school aged children. The coefficients for grades 3 and 4 are negative and statistically significant. These coefficients indicate that schools with high poverty experienced an additional decline of 2.42 and 2.97 points on ELA mean test scores for third and fourth graders, respectively. This result aligns with the hypothesis that younger children were affected the most by the effects of the pandemic. The remainder of the grades have differing outcomes. These results display unexpected and positive outcomes for grades 5-8, with high poverty schools performing better than low poverty schools.

While this model performs as expected, there is a significant amount of room for error and bias. The simplicity of the model fails to consider variations in the poverty rate and how these fluctuations impact ELA comprehension. Since schools have been grouped into two categories defining poverty rate, the intensity of each school's individual poverty rate has been subdued, ultimately leading to the model not capturing the overall effects of the pandemic. Schools within the high poverty category may have experienced differing levels of difficulties, which this model does not encapsulate. Furthermore, this preliminary model fails to consider pre-trends leading up to the pandemic. This exclusion of any pre-trend data has potential for bias that is difficult to determine due to its simplicity.

3.2 Iteration & Improvement

In trying to improve the preliminary model, the most important adjustments were for the binary poverty rate variable and inclusion of further data in the pre-period.

The binary poverty variable caused potential bias and excluded much of the differences in school poverty rates due to all schools being categorized into two groups. This decision resulted in the binary poverty treatment variable to be converted into a continuous treatment variable. This adjustment allows for deeper analysis on the effects of the pandemic for the

variation of poverty rates. Instead of an interaction term for the binary treatment variable and the post-period, the interaction term transitioned into an interaction term between the poverty rate percentage for each school and the post-period. The poverty rate percentage used a baseline value from 2019 for each school. The COVID-19 pandemic had extreme effects on increasing the poverty rate in the United States, especially in 2022 due to the expiration of federal pandemic relief (Trisi). The baseline was used to avoid post-COVID shocks in the poverty rate to provide a consistent metric for each school.

The incorporation of more data led to model adjustments and improvements. Per the codebook for the ELA test score data, the New York City Department of Education determined that the best comparable years are 2018 – 2022. Years outside of this range were incomparable due to a change in testing procedures and testing scale. To incorporate more post-period information, the 2023 test scores were scaled using z-scores and comparing against the mean citywide average. This was unsuccessful due to the arbitrary scale used for that year of testing scores. Nevertheless, the incorporation of information from 2018 was successful due to similar scaling.

With the inclusion of information from 2018, the model required more preliminary assessments and data validation to verify the correct performance. Validating the parallel trends assumption was a critical part of the new model performance. As discussed, the parallel trends assumption is difficult to evaluate and is never truly perfect. This resulted in further analysis to verify the parallel trends assumption using a method that does not restrict the validation to a binary pass or fail. The results of the robustness check were used to interpret coefficients and identify bias in the improved model.

Model	Years	Parallel Trends Assumption Checked?	Treatment Type	Poverty Rate
Original Model	2019, 2022	No, not necessary.	Binary Poverty Variable (1: High Poverty School, 0: Low Poverty School), threshold >60%	2019 Poverty Rate as baseline
Improved Model	2018, 2019, 2022	Yes.	Continuous Poverty Variable	2019 Poverty Rate as baseline

Table 2: Model Version Comparison

Table 2 displays the essential differences between the initial model and the improved model. These changes in the improved model allow for better implementation, improved accuracy, and more valuable results.

3.3 Final Model & Results

The final model uses all possible comparable years to assess pre and post trends in mean school ELA scores. The statistical equation for the improved model can be referenced in Section 2.3 as *Equation 2*. This model uses continuous treatment, in which case is the poverty rate percentage for individual schools using information from 2019 as a baseline. The results from the improved model can be viewed in *Table 3*.

Grade	Pre-trend Coefficient (β_1)	Post-trend Coefficient (β_2)	P-value (post)
3	-0.0231	-0.0453	0.0000
4	-0.0288	-0.0684	0.0000
5	0.0057	0.0207	0.0027
6	0.0394	0.1006	0.0000
7	0.0417	0.0930	0.0000
8	-0.0014	0.0047	0.7571

Table 3: Improved Model Results

The model results align with the initial expectations of younger grades being affected the most by the pandemic. Grades 3 and 4 reported significant learning difficulties in the post-COVID period. The interaction coefficient (β_2) for both third and fourth graders is negative and statistically significant. For results in grade 3, for one percentage point increase in a school's 2019 baseline poverty rate, the post-COVID mean scale score for ELA exams for third graders in an individual school declined by an additional 0.0453 points. For results in grade 4, for one percentage point increase in the poverty rate, the post-COVID mean scale score for fourth graders declined by 0.0684 points. These findings conclude that negative educational outcomes experienced by schools with high poverty rates were exacerbated by the pandemic.

Grade	Magnitude (M)	Conclusion
3	1	✓
4	1.5	✓
5	0	✗
6	0	✗
7	0	✗
8	0	✗

Table 4: HonestDiD Sensitivity Analysis Results

The robustness of the models was validated using the HonestDiD framework to perform a sensitivity analysis and investigate potential biased coefficients based on the parallel trends assumption. Due to the limitations of one pre-trend benchmark difference (2018 – 2019) and one post-period (2022), there is only one benchmark difference to compare. In Table 4, the results for the robustness check are displayed for all models. The results show that the findings for post-COVID effects still hold despite the violation of the parallel trends assumption for third and fourth grades. The model breaks when magnitudes for third and fourth graders exceed 1 and 1.5,

respectively, in which the model breaks when larger violations occur. Unfortunately, for grades 5-8, the magnitude is reported as 0. The sensitivity analysis identifies that the models were not statistically significant, even under the assumption of an unrealistic, perfect parallel trends. The robustness check validates the results, but the coefficients are not completely undeniable.

4. Conclusion & Future Directions

This project aimed to quantify learning outcomes for impoverished schools in the post-pandemic period relative to pre-trends before the pandemic. Using a two-way fixed effects difference-in-difference model, this research explored the achievement gap between impoverished and non-impoverished students and sought to explore not only why students fell behind, but also by how much. Using data from the New York City Department of Education, schools' average ELA test scores were evaluated to quantify learning gaps and differences in outcomes. Models were performed separately by grade to identify which grades were affected the most. Results were somewhat successful, with findings showing that third and fourth grade students experienced learning difficulties in the post-pandemic period, as schools with higher baseline poverty rates in the pre-pandemic period experienced larger declines in ELA test scores in 2022.

Working with data surrounding the COVID-19 pandemic poses unique challenges with data structure and consistency in information. With this data in particular, ELA exams were not administered in 2020 and optional in 2021. Due to this inconsistency, the research lacked a significant portion of data for some of the most critical years. Nevertheless, the time jump from 2019 to 2022 allowed for insightful analysis of pre-trends and post-COVID recovery outcomes.

This project strictly worked with data from New York City, which is a unique and complex school system in one of the largest cities in the world. These results are unique to the New York City school system and may not be aligned with the overall effects of the pandemic on impoverished schools across the nation. School systems in rural, sparsely populated communities may yield different results due to the differences in educational infrastructure and available resources.

An interesting area for future research would be further analysis of test score comprehension levels, as defined by the NYC DOE. An investigation of trends in ELA proficiency levels would provide a more insightful evaluation of how student performance shifted following the pandemic.

The findings of this research highlight the importance of post-pandemic educational recovery, especially for marginalized communities. Understanding how poverty influences post-pandemic education outcomes is essential for policymakers and educators to make informed decisions about resource allocation and recovery programs. These findings reinforce the need for equitable interventions that support students that are most affected by academic disruptions.

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Appendix

Table 1A: Count of Grade Level Observations per Year

<i>Grade</i>	<i>Years</i>	<i>Count</i>
3	2018	764
	2019	774
	2022	776
4	2018	750
	2019	764
	2022	770
5	2018	743
	2019	756
	2022	775
6	2018	474
	2019	476
	2022	474
7	2018	466
	2019	467
	2022	469
8	2018	466
	2019	464
	2022	465

Table 2A: Summary Statistics of Mean Scale Score, Number of Students Tested, and Poverty Rate

	<i>Year</i>	<i>Mean</i>	<i>Standard Deviation</i>	<i>Min.</i>	<i>Max.</i>
<i>Mean Scale Score</i>	2018	600.549	9.982	558	639
	2019	599.727	10.143	559	644
	2022	600.061	10.583	567	635
<i>Number Tested</i>	2018	102.776	86.004	6	745
	2019	101.322	86.498	6	737
	2022	82.662	74.930	6	696
<i>Poverty Rate (%)</i>	2018	76.035	21.822	5.6	95
	2019	74.805	21.796	7.2	95
	2022	74.376	21.198	6.1	95

Figure A1: Changes in ELA Score by School District

